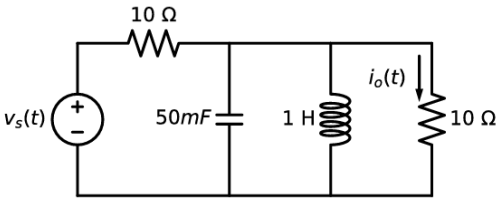
Escuela Politécnica Nacional

Análisis de Circuitos Eléctricos

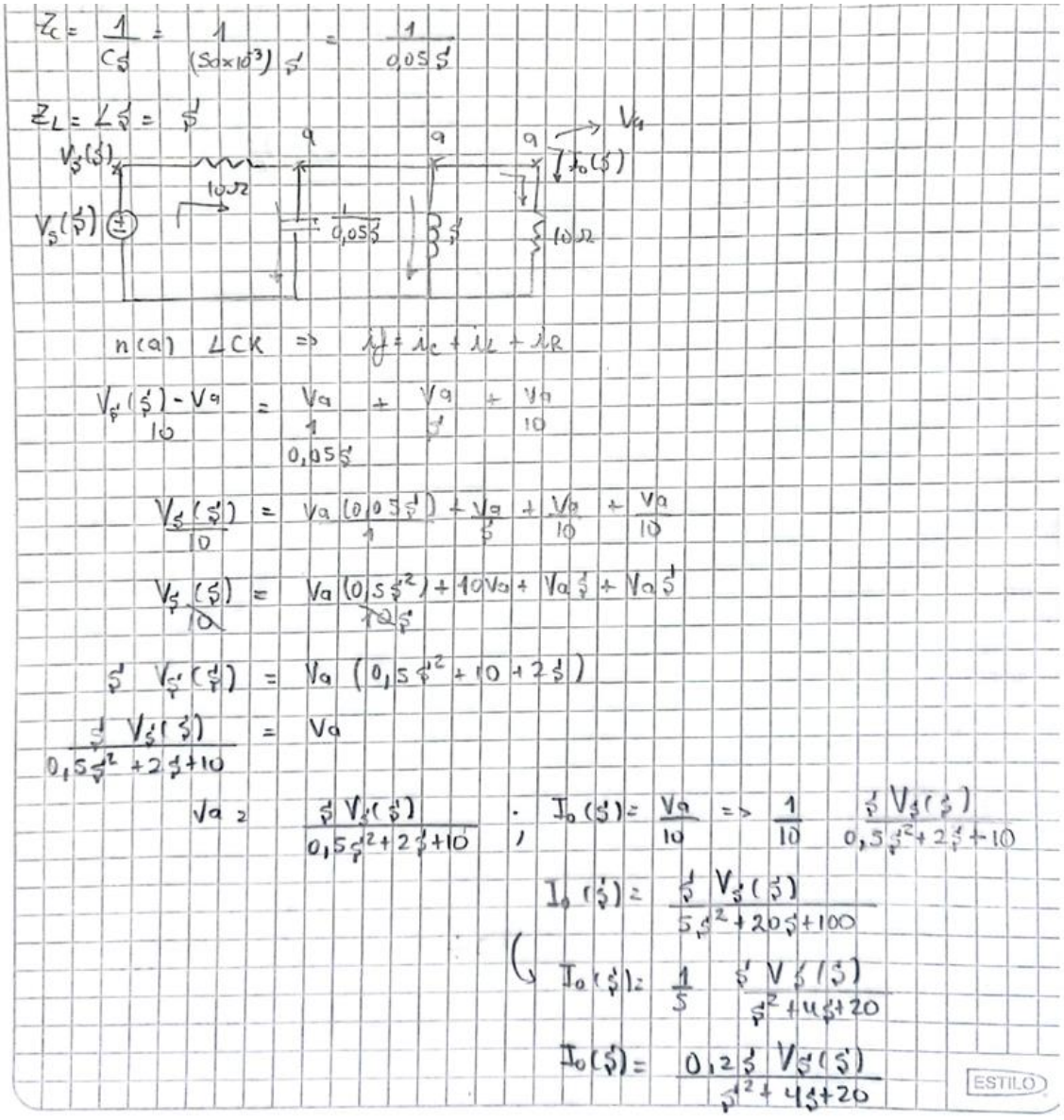
Evaluación 3 -GR3

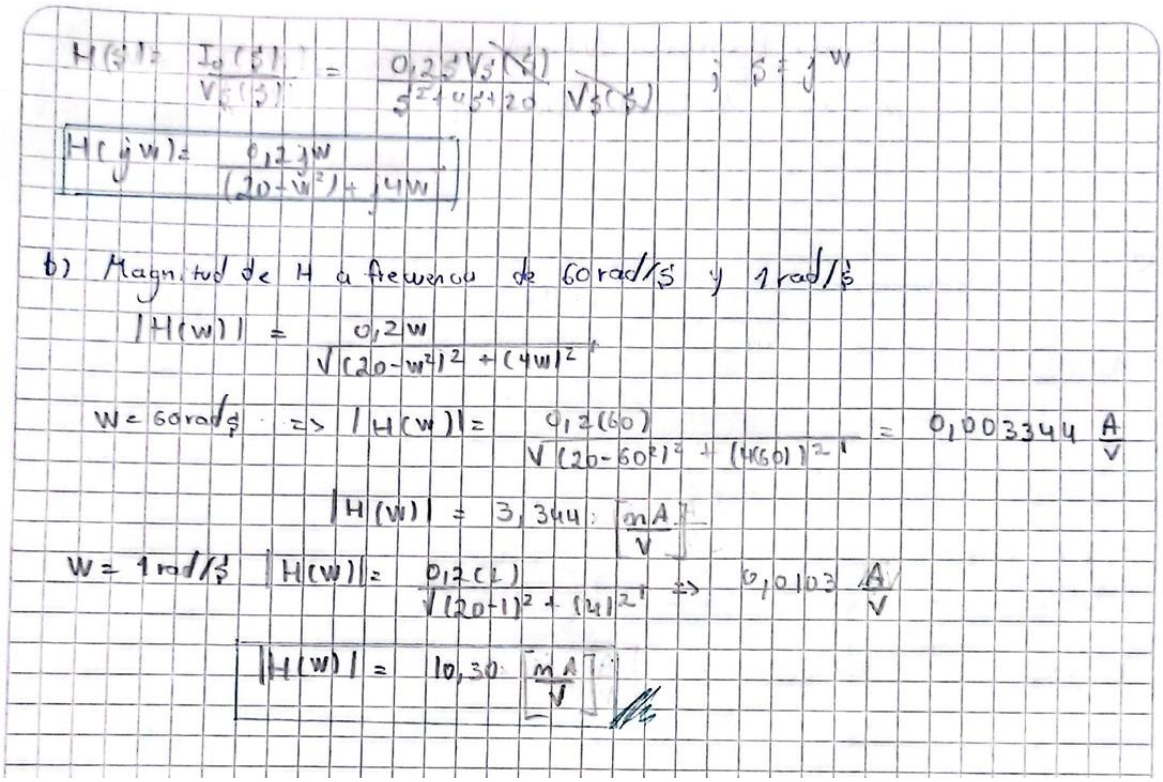
Nombre: Jhon Durán

NOMBRE: .....



1. Para el siguiente circuito determine:
   a. La función de red $H(\omega) = I_0(\omega)/V_s(\omega)$ (**1 punto**)
   b. La magnitud de H una frecuencia de 60 rad/s y 1 rad/s (**1.5 puntos**)





2. El circuito de la Figura 1, se encuentra en régimen permanente. En el instante $t = 0$ se cierra el interruptor S_1 y se abre S_2. Calcular el voltaje $v_C(t)$ para $t > 0$ usando la transformada de Laplace (**4 puntos**).

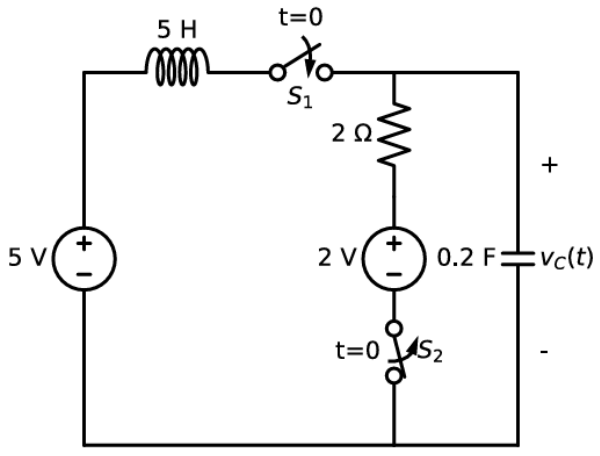
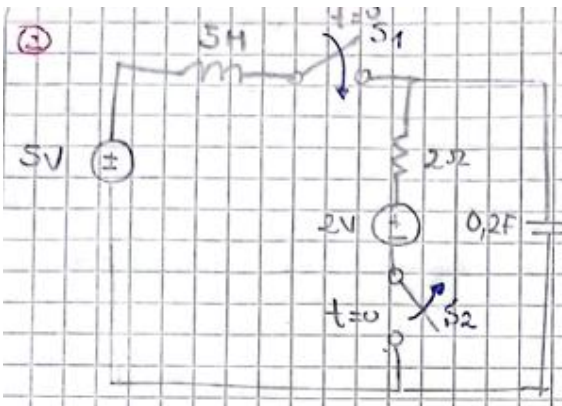


Figura 1

②

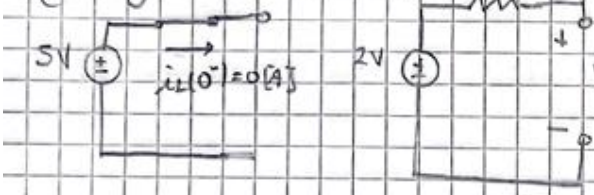


Condiciones Iniciales

$i_L(0^-) = i_L(0^+) = 0\,[A]$

$V_c(0^-) = 2\,[V]$ $\quad V_c(0^-) = V_c(0^+) = 2\,[V]$

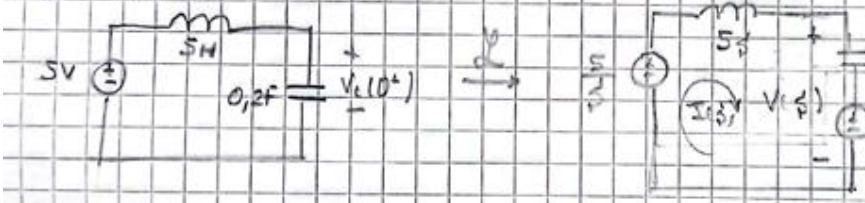
Analizando en $t=0^-$



$\frac{1}{Cs} = \frac{1}{0,2s} \Rightarrow \frac{5}{s}$

$\frac{V(0)}{s} = \frac{2}{s}$

en $t>0$



Aplicando LVK en el circuito en el dominio de s:

M(I)

$$-\frac{2}{s} + 5s\,I(s) + \frac{5}{s}\,I(s) + \frac{2}{s} = 0$$

$$I(s)\left(5s + \frac{5}{s}\right) = \frac{5}{s} - \frac{2}{s} \Rightarrow I(s)\left(\frac{5s^2+5}{s}\right) = \frac{3}{s}$$

$$I(s) = \frac{3}{5(s^2+1)} \quad ; \quad V_c(s) = I(s)\frac{5}{s} + \frac{2}{s}$$

$$V_c(s) = \frac{3}{5(s^2+1)}\cdot\frac{5}{s} + \frac{2}{s}$$

$$V_c(s) = \frac{3}{s(s^2+1)} + \frac{2}{s}$$

$$V_c(s) = \frac{3 + 2s^2 + 2}{s(s^2+1)}$$

$$V_c(s) = \frac{3 + 2s^2 + 2}{s(s^2+1)} \Rightarrow V_c(s) = \frac{2s^2+5}{s(s^2+1)}$$

ESTILO

$$V_c(s) = \frac{2s^2+5}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1} = \frac{As^2+A+Bs^2+Cs}{s(s^2+1)}$$

$$\begin{cases} A+B=2 \\ A=5 \\ C=0 \end{cases} \Rightarrow B=-3$$

$$V_c(s) = \frac{5}{s} - \frac{3s}{s^2+1} \xrightarrow{\mathcal{L}^{-1}} v_c(t) = 5 - 3\cos(t)\ [V], \text{ para } t>0$$

Comprobando:

$v_c(0) = 5 - 3\cos(0)$

$= 5-3$

$v_c(0) = 2\ [V]$ ¿coincide con la C.I.

$\left.\frac{d\,v_c(t)}{dt}\right|_{t=0} = 3\,\text{sen}(t)\Big|_0 \Rightarrow 0$ $i_L(0)=0$ ¿coincide con la C.I

3. En el circuito de la Figura 2. Las condiciones iniciales son: $i_{L1}(0^-) = 0.5\ A, i_{L2}(0^-) = 0\ A$. Calcular $v_R(t)$ para $t > 0$ usando la transformada de Laplace (**3.5 puntos**).

t=0

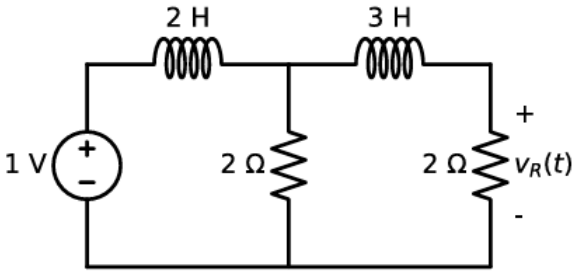
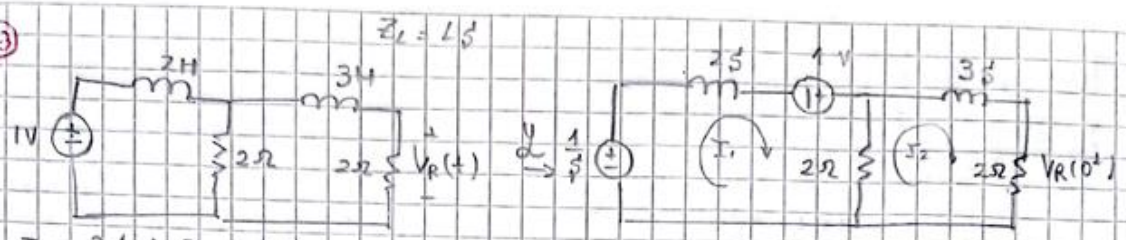


Figura 2

③



$Z_{L_1} = 2\,\mathcal{S}$; $Z_{L_2} = 3\,\mathcal{S}$

$\mathcal{L}i(0) = 2(0{,}5)$

$\mathcal{L}_1 i(0) = 1$

LVK Malla 1

$$-\frac{1}{\mathcal{S}} + 2\mathcal{S}I_1 - 1 + 2(I_1 - I_2) = 0$$

$$-\frac{1}{\mathcal{S}} - 1 + 2\mathcal{S}I_1 + 2I_1 - 2I_2 = 0 \quad (1)$$

LVK Malla 2

$$2(I_2 - I_1) + 3\mathcal{S}I_2 + 2I_2 = 0$$

$$2I_2 - 2I_1 + 3\mathcal{S}I_2 + 2I_2 = 0$$

$$4I_2 + 3\mathcal{S}I_2 = 2I_1$$

$$I_2 = \frac{2I_1}{4 + 3\mathcal{S}} \quad (2)$$

(2) en (1)

$$-\frac{1}{\mathcal{S}} - 1 + 2\mathcal{S}I_1 + 2I_1 - 2\left(\frac{2I_1}{4+3\mathcal{S}}\right) = 0$$

$$-\frac{1-\mathcal{S}}{\mathcal{S}} + 2\mathcal{S}I_1 + 2I_1 - \frac{4I_1}{4+3\mathcal{S}} = 0$$

$$I_1\left(\frac{2\mathcal{S}}{1} + \frac{2}{1} - \frac{4}{4+3\mathcal{S}}\right) = \frac{1+\mathcal{S}}{\mathcal{S}}$$

$$I_1\left(\frac{8\mathcal{S} + 6\mathcal{S}^2 + 8 + 6\mathcal{S} - 4}{4+3\mathcal{S}}\right) = \frac{1+\mathcal{S}}{\mathcal{S}}$$

$$I_1\left(\frac{6\mathcal{S}^2 + 14\mathcal{S} + 4}{4+3\mathcal{S}}\right) = \frac{1+\mathcal{S}}{\mathcal{S}}$$

$$I_1\left(\frac{(3\mathcal{S}+1)(2\mathcal{S}+4)}{4+3\mathcal{S}}\right) = \frac{1+\mathcal{S}}{\mathcal{S}}$$

$$I_1 = \frac{(1+\mathcal{S})(4+3\mathcal{S})}{(3\mathcal{S}+1)(2\mathcal{S}+4)(\mathcal{S})}$$

$6\mathcal{S}^2 + 14\mathcal{S} + 4$

3×1

2×4

ESTILO

Reemplazando I₁ en (2)

$$I_2 = \frac{2 I_1}{4+3s}$$

$$= \frac{2}{4+3s} \cdot \frac{(1+s)(4+3s)}{s(3s+1)(s+2)}$$

$$I_2 = \frac{s+1}{s(3s+1)(s+2)} = \frac{A}{s} + \frac{B}{3s+1} + \frac{C}{s+2}$$

$$= \frac{A(3s+1)(s+2) + B(s)(s+2) + C(s)(3s+1)}{s(3s+1)(s+2)}$$

$$s+1 = A(3s^2+6s+s+2) + B(s^2+2s) + C(3s^2+s)$$

$$s+1 = (3A+B+3C)s^2 + (7A+2B+C)s + 2A$$

$$\begin{cases} 3A+B+3C = 0 \\ 7A+2B+C = 1 \\ 2A = 1 \end{cases}$$

Resolviendo sistema.

$$A = \frac{1}{2} \quad ; \quad B = -\frac{6}{5} \quad C = -\frac{1}{10}$$

$$I_2(s) = \frac{(1/2)}{s} - \frac{(6/5)}{3s+1} - \frac{1/10}{s+2}$$

$$\frac{6}{5} \cdot \frac{1}{3\left(s+\frac{1}{3}\right)} = \frac{2}{5} e^{(-1/3)t} \qquad \frac{6}{5} \quad \frac{1}{3s+1}$$

$$i_{12}(t) = \frac{1}{2} - \frac{2}{5} e^{-\frac{1}{3}t} - \frac{1}{10} e^{-2t} \ [A] \quad \text{para } t>0$$

$$V_{R_2}(t) = i_2(t) \cdot 2 \Rightarrow 1 - \frac{4}{5} e^{(-1/3)t} - \frac{1}{5} e^{-2t} \ [A] \ , \ t>0$$

$$\boxed{V_{R_2}(t) = 1 - 0{,}8\, e^{-\frac{1}{3}t} - 0{,}2\, e^{-2t} \ [V] \ , \ \text{para } t>0}$$